Canadian Journal of
Fisheries and
Aquatic Sciences

Research Article

# Dissolved organic carbon enhances terrestrial carbon pathways with limited effects on freshwater fish trophic position

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## Abstract

Freshwater ecosystems are affected by fluctuations in terrestrially derived dissolved organic carbon (DOC). Increased DOC runoff from watersheds can contribute to the "browning" of freshwater systems, altering water properties and disrupting food web dynamics, with potential impacts on fish. This study examined how carbon sources and the trophic ecology of temperate freshwater fishes varied with DOC concentrations. We used stable isotope ratios of carbon ($\delta^{13}$C) and nitrogen ($\delta^{15}$N) to trace carbon sources in organisms at low trophic levels and to assess the trophic ecology of six fish species from 70 lakes in Sweden, Germany, Canada, and the United States. We found that both pelagic and benthic $\delta^{13}$C decreased as DOC increased, suggesting higher reliance on terrestrial carbon by lower trophic level organisms. Fish responses to elevated DOC were limited to certain species. Mid-trophic level European perch (*Perca fluviatilis*) increased pelagic diet proportions, while top predators like northern pike (*Esox Lucius*) and walleye (*Sander vitreus*) showed nonlinear shifts. Except for walleye, no effect of DOC was detected on fish trophic position. Our findings suggest that DOC alters energy use and diet but not trophic position in some freshwater fish.

**Key words:** freshwater ecosystems, food web dynamics, fish, interspecific variation, stable isotope analysis, browning

## Introduction

Freshwater ecosystems face continuous challenges from anthropogenic stressors (Woodward et al. 2010; Weiskopf et al. 2020). Warmer temperatures and increased precipitation that accompany global climate change can contribute to elevated levels of terrestrially derived dissolved organic carbon (DOC) in freshwater systems (Larsen et al. 2011). This increase in DOC, driven by changes in soil ionic strength and pH resulting from shifting atmospheric deposition (De Wit et al. 2021), leads to the darkening of lakes and rivers in many regions worldwide—a phenomenon commonly referred to as "browning" (Monteith et al. 2007; Meyer-Jacob et al. 2019; Kritzberg et al. 2020). Although an increase in water colour is commonly explained by an increase in DOC, the colour of water is also related to the quality of DOC and the prevalence of iron (Kritzberg and Ekström 2012). The influx of terrestrially derived (i.e., allochthonous) DOC to freshwater ecosystems is concerning as it can impact both chemical and physical properties of freshwaters, and influence food web structure and dynamics (Solomon et al. 2015; Creed et al. 2018). Such changes can have cascading effects on fish populations, which are crucial for recreational, domestic, and commercial fisheries (Karlsson et al. 2009; van Dorst et al. 2019). Thus, understanding how changes in DOC concentrations affect food webs at a broad spatial scale is essential for effective freshwater ecosystem management.

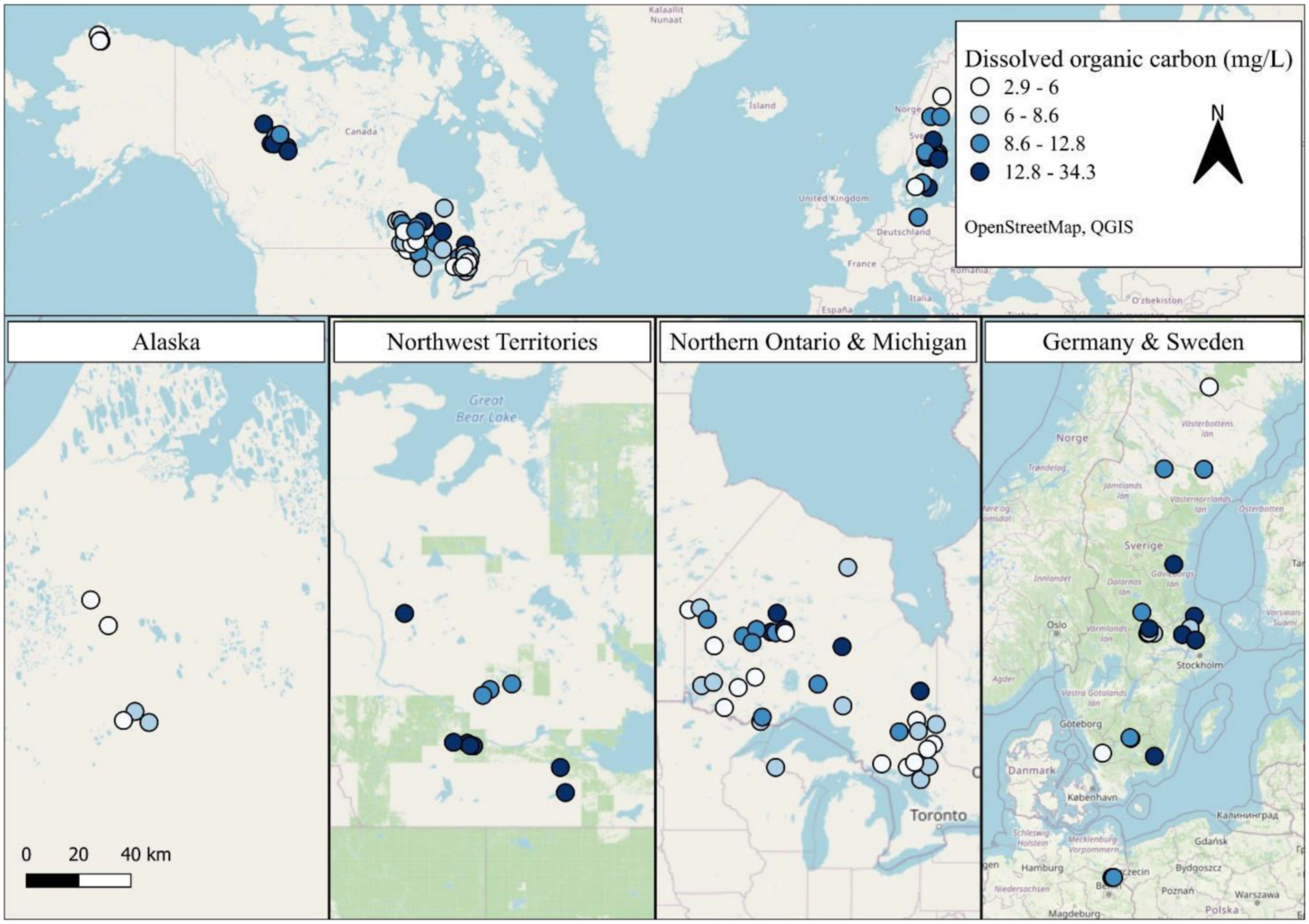
Increases in allochthonous DOC can modify carbon sources and energy pathways in freshwater ecosystems (Brett et al. 2017; Creed et al. 2018). Terrestrial derived DOC from watersheds can enhance pelagic primary production (i.e., phytoplankton) in oligotrophic lakes by supplying additional nutrients (Ask et al. 2012; Seekell et al. 2015; Solomon et al. 2015), though it may concurrently reduce benthic primary production by increasing shading (Seekell et al. 2015). However, as DOC concentrations become even further elevated, pelagic production may eventually decrease due to limited light availability for phytoplankton growth (Karlsson et al. 2009; Finstad et al. 2014). Alternatively, increased levels of terrestrially derived DOC can stimulate microbial activity, fuelling both benthic and pelagic pathways to higher trophic levels, resulting in a trophic upsurge (Carpenter et al. 2005; Solomon et al. 2011; Tanentzap et al. 2014). For example, Robbins et al. (2020) found that increased terrestrial DOC enhanced periphyton bacterial production and biomass, leading to higher benthic invertebrate abundance in boreal streams. Studies suggest that there is no single critical DOC concentration or threshold beyond which production declines, but rather one that varies systematically from lake to lake depending on lake-specific characteristics (Finstad et al. 2014; Kelly et al. 2018; Jane et al. 2024). Consequently, DOC concentrations play a pivotal and complex role in shaping both carbon dynamics and energy flows within freshwater food webs. Spatial variation in the causes and extent of changes in DOC adds further complexity to understanding and predicting how food webs will respond (Blanchet et al. 2022). Investigating responses of food webs to increases in DOC across environmental gradients is required to further elucidate the broader ecological impacts of rising DOC levels in freshwater ecosystems.

Most studies assessing the impacts of increased DOC concentrations in freshwater have focused on primary consumers, such as zooplankton (Tanentzap et al. 2017; Creed et al. 2018; Eie et al. 2021).While a majority of research has focused on lower trophic levels of aquatic food webs, there is an increasing body of work examining the impacts of elevated DOC on top consumers, including fish. Existing research suggests that increased DOC and water browning (which can occur with increased DOC, depending on the quality) can affect fish diets directly by reducing visibility and thus prey detectability, and indirectly by altering primary production and prey abundance and composition (Hedström et al. 2017; Koizumi et al. 2018; Leech et al. 2021). For instance, Bartels et al. (2016) found that higher DOC concentrations in lakes caused European perch (*Perca fluviatilis*) to rely more on pelagic resources. However, fish responses to increased DOC and browning may vary based on species-specific feeding and foraging strategies (Estlander et al. 2010; Weidel et al. 2017). For example, Stasko et al. (2015) found that generalist smallmouth bass (*Micropterus dolomieu*) increasingly relied on pelagic resources and exhibited a wider trophic range as water clarity decreased, whereas walleye (*Sander vitreus*)—a species adapted to low-light conditions—showed no dietary change. Dietary shifts in fish due to increased DOC could lead to changes in their trophic position, as organisms supported by pelagic pathways are generally smaller and occupy lower trophic levels (Maitland et al. 2024). Understanding interspecific differences is essential for predicting species vulnerability to changes in DOC and the browning of freshwater ecosystems, as well as anticipating long-term impacts, such as shifts in fish growth (van Dorst et al. 2020) and production (Karlsson et al. 2015; van Dorst et al. 2019).

Stable isotopes ($^{13}$C, $^{15}$N) are useful tools for tracing carbon sources and understanding food web ecology (Post 2002), but their use in studying how increased DOC affects food web dynamics and fish trophic ecology remains limited (e.g., Solomon et al. 2011; Bartels et al. 2012; Stasko et al. 2015; Hayden et al. 2019). Ratios of $^{13}$C ($\delta^{13}$C) are particularly useful for distinguishing carbon sources in freshwater systems. Typically, pelagic (autochthonous) producers exhibit more depleted (more negative) $\delta^{13}$C values, ranging from −42 to −26 ‰, compared to benthic producers, which tend to have more enriched (less negative) $\delta^{13}$C values, ranging −30 to −12‰ (France 1995; Post 2002). However, there is some overlap in these ranges. Additionally, $\delta^{13}$C values can help assess terrestrial derived (allochthonous) carbon inputs, as these sources show more depleted $\delta^{13}$C values, typically around −28 to −30 ‰, though this range may vary depending on the type of vegetation within the watershed (Peterson and Fry 1987; Solomon et al. 2011; Tanentzap et al. 2014; Eglite et al. 2024). Stable isotope analysis can reveal fish resource use and trophic position, making it ideal for investigating how DOC influences interspecific variation in fish diets.

In this study, we explored the effects of DOC concentrations on lake food web structure across a broad spatial scale. Specifically, we aimed to (1) investigate how DOC concentrations affect carbon sources and energy flow within the food web, and (2) assess how DOC concentrations influence interspecific variation in the importance (i.e., proportion) of pelagically sourced carbon and trophic position in fish. We used $\delta^{13}$C and $\delta^{15}$N to infer carbon sources and food web positions, respectively, of pelagic and benthic organisms at low trophic levels and to quantify the trophic ecology of six fish species varying in foraging strategies. Our study regrouped a large dataset from 70 lakes spanning a wide DOC gradient (2.4–34.3 mg/L) at a broad spatial scale in regions across northern Europe and North America: Sweden, Germany, Northern Ontario (Canada), Northwest Territories (Canada), Michigan (United States), and Alaska (United States). For objective 1, we hypothesized that increased DOC would enhance the incorporation of allochthonous carbon into both pelagic and benthic pathways, consistent with the trophic upsurge hypothesis (Tanentzap et al. 2014; Robbins et al. 2020), and predicted a negative correlation between baseline $\delta^{13}$C values and lake DOC concentrations. For objective 2, if the influx of terrestrial carbon stimulates productivity at lower trophic levels, we expected species-dependent cascading effects on fish populations (Stasko et al. 2015; Bartels et al. 2016). We anticipated that mid-trophic level fish would respond to this change by increasing their proportional consumption of small pelagic prey such as zooplankton, in relation to higher trophic level benthic prey (e.g., macroinvertebrates or benthic fish). On the

**Fig. 1.** Map showing the distribution of studied lakes across a broad spatial scale, including Alaska, United States ($n = 5$), Northwest Territories, Canada ($n = 10$), Northern Ontario, Canada ($n = 35$), Michigan, United States ($n = 1$), Germany ($n = 2$), and Sweden ($n = 17$). Lakes are colour-coded along a gradient of dissolved organic carbon concentration (DOC, mg/L). The map uses the WGS 84/Pseudo-Mercator projection, and original latitude and longitude coordinates were reprojected to align with the basemap.




other hand, if there is no dietary change for mid-trophic level fish or top predator fish in response to browning, and they continued feeding on similar proportions of fish prey, then their trophic position would remain unchanged. This prediction is supported by studies showing limited dietary flexibility in top predators under changing conditions (e.g., Rush et al. 2012). Ultimately, our study enhances understanding of how ongoing environmental changes associated with altered DOC concentrations in lakes influence freshwater food web dynamics at a broad spatial scale.

## Materials and methods

### Study area

Our study investigated 70 lakes from across Europe, Canada, and the United States across a wide range of DOC concentrations. In Europe, we sampled lakes in Sweden ($n = 17$), and Germany ($n = 2$), where DOC concentrations ranged from 3.6 to 34.3 mg/L. In Canada, our sampling sites included lakes in Northern Ontario ($n = 35$), and the Northwest Territories ($n = 10$), while in the United States, we sampled lakes in the states of Michigan ($n = 1$), and Alaska ($n = 5$), with DOC concentrations ranging from 2.4 to 21.3 mg/L (Fig. 1). The lakes were selected based on availability of stable isotope data from fish with corresponding baseline data. Our selection aimed to capture a broad range of DOC concentrations across a large spatial scale. From the datasets, we selected European perch, yellow perch (*Perca flavescens*), northern pike (*Esox lucius*), walleye, smallmouth bass, largemouth bass (*Micropterus salmoides*), and ninespine stickleback (*Pungitius pungitius*) as focal study species due to their prevalence in the sampled lakes, the availability of stable isotope data, and their representation of diverse roles in the food web—mid-trophic level fish and top predators. We combined smallmouth bass and largemouth bass data into a single category, referred to as "black bass", due to the limited availability of largemouth bass data (from only one lake) and the ecological similarities between the two species (Philipp and Ridgway 2003). A summary of the data used in this study is provided in Table 1.

**Table 1.** Summary of the data used in the study.

| Region | Lakes ($n$) | Total phosphorus range (μg/L) | DOC range (mg/L) | Fish species | Fish length (mm)[a] | Benthic baseline | Pelagic baseline | Source |
|---|---|---|---|---|---|---|---|---|
| Sweden | 17 | 3.8–46.7 | 3.6–34.3 | European perch ($n$ = 820) | 133 (39–388) | Benthic macro-invertebrates | Zooplankton | Bartels et al. (2016); Andersson (2021) |
| Germany | 2 | 4.4–27.0 | 10.3–12 | European perch ($n$ = 105) | 153 (58–338) | Benthic macro-invertebrates | Zooplankton | Bartels et al. (2016); Andersson (2021) |
| Northern Ontario | 35 | 3.7–20.3 | 2.4–16.2 | Northern pike ($n$ = 410) Walleye ($n$ = 686) Yellow perch ($n$ = 519) Smallmouth bass ($n$ = 239) | Northern Pike: 585 (115–1066) Walleye: 429 (89–815) Yellow perch: 99 (25–290) Smallmouth bass: 352 (70–518) | Gastropoda ($n$ = 32), plecoptera ($n$ = 2), periphyton ($n$ = 1) | Bivalvia ($n$ = 34), zooplankton ($n$ = 1) | Johnston et al. (2024); Tanentzap et al. (2014) |
| Northwest Territories | 10 | 7.0–30.5 | 9.8–21.3 | Northern pike ($n$ = 422) | 565 (149–1042) | Gastropoda | Bivalvia | Moslemi-Aqdam et al. (2022) |
| Michigan | 1 | 16.9–17.7 | 7.9 | Largemouth bass ($n$ = 16) | 191 (58–378) | Benthic macro-invertebrates | Zooplankton | Solomon et al. (2018) |
| Alaska | 5 | 3.0–11.0 | 4.2–8.1 | Ninespine stickleback ($n$ = 195) | 53 (31–73) | Gastropoda | Bivalvia | Burke et al. (2020) |

[a]Fish average length is presented and followed by the range in parentheses.

## Field collections and laboratory analyses

Detailed descriptions of field collections and laboratory analyses for each dataset can be found in the following: (Kelly et al. 2014; Tanentzap et al. 2014; Craig et al. 2015; Bartels et al. 2016; Koizumi et al. 2018; Solomon et al. 2018; Burke et al. 2020; Andersson 2021; Moslemi-Aqdam et al. 2021; Johnston et al. 2024). In general, surface water was collected for DOC and total phosphorus analyses, while fish, macroinvertebrates, and zooplankton were collected for stable isotope analysis. Macroinvertebrates were sampled using a variety of methods, including kick-net sampling and hand-picking, while zooplankton was collected using nets of varying mesh sizes. In Sweden, sampling for macroinvertebrates was done in the littoral zone, and for zooplankton, in the pelagic zone. In Northern Ontario, macroinvertebrates and bivalves were sampled from shallow depositional zones. In the Alaska region, macroinvertebrates, including bivalves, were collected from both nearshore and offshore locations; however, the lakes were small (0.5–2.5 $km^2$) and shallow (2-4 m). In the Northwest Territories, benthic invertebrates were sampled from both nearshore and offshore areas. Fish were collected using a combination of gillnets, fyke nets, and electrofishing. In the laboratory, fish were dissected to obtain white dorsal muscle tissue, or, in the case of smaller fish, the entire specimen was used and frozen at −20 °C. Whole-body samples were used only for a subset of the yellow perch data from Northern Ontario (Tanentzap et al. 2014). For stable isotope analysis, only the soft tissues of gastropods and bivalves were retained. The stable isotopic ratios of carbon ($\delta^{13}C$) and nitrogen ($\delta^{15}N$) were measured on dried, homogenized samples of all biota using isotope ratio mass spectrometers at various laboratories. Stable isotope ratios (‰) were calculated by comparing the ratio of heavy to light isotopes in the sample to the ratio in international standards: Vienna Peedee Belemnite (for carbon) and atmospheric nitrogen ($N_2$; for nitrogen). These ratios were calculated using the following equation:

$$\delta R\,(‰) = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1\right) \times 1000$$

where $R$ is the ratio $^{15}N$:$^{14}N$ or $^{13}C$:$^{12}C$ (Fry 2006). Instrumental precision based on replicate measurements varied between 0.02 and 0.3‰ for $\delta^{15}N$ and 0.05 and 0.4‰ for $\delta^{13}C$, depending on the laboratory. Specifically, for data from Johnston et al. (2024), the mean coefficients of variation for duplicate analyses of the sample material were 0.20% for $\delta^{13}C$ and 0.89% for $\delta^{15}N$. We applied the Kiljunen et al. (2006) lipid-normalizing model to adjust the $\delta^{13}C$ for all fish species in the dataset (isotopic difference between protein and lipid of 7.018 ‰ and a constant of 0.048). This adjustment was necessary because some individuals within species (e.g., ninespine stickleback, walleye, and smallmouth bass) had a C/N ratio greater than 4, and to ensure consistency across the dataset, we standardized the $\delta^{13}C$ for all fish species. Raw isotopic data for baseline and fish samples are presented in Figs. S1 and S2, respectively. After evaluating the isotopic signals from both tissue types for yellow perch while accounting for baseline effects, we found that both exhibited a similar range of $\delta^{13}C$ values (Fig. S3). This is consistent with findings from Eglite



Can. J. Fish. Aquat. Sci. **82:** 1–15 (2025) | dx.doi.org/10.1139/cjfas-2024-0343

et al. (2024) who reported small isotopic differences between yellow perch tissue in Lake Michigan.

## Animal ethics

Fish sampling protocols for Northern Ontario lakes were reviewed and approved by the Ministry of Natural Resources' Fisheries Animal Care Committee (FACC-50, ARMS-ACC-97, ARMS-ACC-97B) and conducted in accordance with Laurentian University's Animal Care Certificate (2013-03-03). Sampling in the Northwest Territories was performed under Aurora Research Institute Scientific Research Licenses (#16046 and #16633) and adhered to the University of Waterloo Animal Care Committee's Animal Use Protocols (40576 and A-18-04). In Alaska, fish collection was authorized by a U.S. government permit, with handling procedures following the University of Waterloo Animal Care Committee's Protocol (A-13-11). Sampling in Michigan lakes was conducted under a Scientific Collector's Permit from the Michigan Department of Natural Resources, with animal care methods approved by the Macdonald Campus Facility Animal Care Committee at McGill University. For Northern Ontario, the Northwest Territories, Alaska, and Michigan, fish sampling complied with Canadian Council on Animal Care guidelines. Sampling in Sweden and Germany was carried out in collaboration with monitoring teams from the National Environmental Monitoring Program and the Integrated Studies of the Effect of Liming Acidified Waters in Sweden, and with the Leibniz Institute of Freshwater Ecology and Inland Fisheries, in Germany. The Swedish Agency for Marine and Water Management supported the standard fish sampling in Swedish lakes. Fish collection in Sweden adhered to Swedish Animal Care Guidelines, with oversight and approval from the Uppsala Authority for the Ethics of Animal Experimentations (license C1/6).

## Data curation

The Northern Ontario lake dataset was extensive, and we employed a systematic approach to handling DOC values. If a single DOC value was provided, we used that value. Where DOC data spanned multiple years, but they did not coincide with the year of fish collection, we calculated the average of available values. However, if a DOC value for the collection year was available, we used that value. Where multiple DOC values were accessible for the collection year, we prioritized the value corresponding to the collection date. If this specific value was unavailable, we averaged the available data. For all other datasets, the DOC value was either a single data point for the lake or the mean average of DOC values matching the year of fish collection.

The stable isotope baselines for both $\delta^{13}$C and $\delta^{15}$N in benthic habitats were represented by macroinvertebrates, such as gastropods, which reflect benthic or littoral primary production through grazing on periphyton and detritus (Post 2002). In pelagic habitats, baselines were represented by zooplankton, which capture the isotopic signature of the pelagic food web (Post 2002). Where zooplankton data were unavailable, we used bivalves as an alternative baseline due to their sessile filter-feeding behaviour, which integrates seston isotopic signals, capturing both spatial and temporal variability (Post 2002). The isotopic signatures of bivalves were consistent across regions and closely aligned with those of zooplankton (Fig. S2). There was a clear isotopic difference between benthic and pelagic baselines within regions, confirming that the selected baselines effectively represented the studied food web pathways (Fig. S2).

To represent each lake, we used lake-specific baseline values for $\delta^{13}$C and $\delta^{15}$N. When multiple baseline samples were available for a given lake, we calculated a mean for the benthic and pelagic baselines per lake. If only a single baseline value for each lake was available, we used that value. Baseline data typically matched the fish collection year; if not, the average of the available years for that lake was used. Each individual fish was therefore paired with baselines from the same lake and, when possible, the same year.

## Fish pelagic diet proportion and trophic position calculations

The proportion of fish diet derived from pelagic carbon sources ($\alpha$), as well as trophic position (TP), were calculated using two-source mixing models from Post (2002):

$$\alpha = \frac{\delta^{13}\mathrm{C}_{\mathrm{consumer}} - \delta^{13}\mathrm{C}_{\mathrm{benthic}}}{\delta^{13}\mathrm{C}_{\mathrm{pelagic}} - \delta^{13}\mathrm{C}_{\mathrm{benthic}}}$$

$$\mathrm{TP} = \lambda + \frac{\delta^{15}\mathrm{N}_{\mathrm{consumer}} - \left[\delta^{15}\mathrm{N}_{\mathrm{pelagic}} \times \alpha + \delta^{15}\mathrm{N}_{\mathrm{benthic}} \times (1-\alpha)\right]}{\Delta n}$$

where $\alpha$ represents the relative contribution of the pelagic baseline to the consumer, and TP was calculated using pelagic and benthic baselines representing primary consumers ($\lambda = 2$). An exception was made for the yellow perch trophic position data from lakes reported by Tanentzap et al. (2014), because pelagic and benthic baselines were, respectively, primary consumers and primary producers and had a different lambda. This mismatch could not be accommodated by the two-source mixing model. As a result, we calculated the trophic position of the yellow perch in Tanentzap et al. (2014) using the single-source equation (Post 2002):

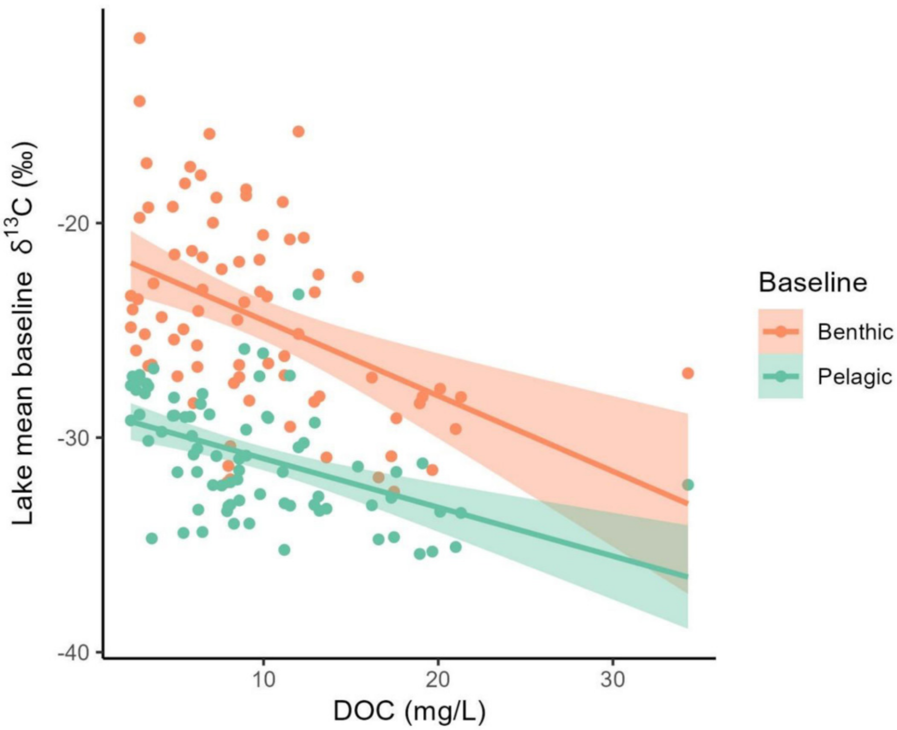
$$\mathrm{TP} = \lambda + \frac{\delta^{15}\mathrm{N}_{\mathrm{consumer}} - \delta^{15}\mathrm{N}_{\mathrm{pelagic}}}{\Delta n}$$

In this case, zooplankton was used as the pelagic baseline, with $\lambda = 2$. Calculations used the nitrogen stable isotope value of the consumer and a trophic discrimination factor for nitrogen ($\Delta$N) of 3.4‰ (Post 2002). Trophic position was used to classify fish species into two main groups: top predator fish and mid-trophic level fish. Yellow perch, European perch, black bass, and ninespine stickleback, with average trophic positions ranging from 3.5 to 4, were categorized as mid-trophic level fish. In contrast, northern pike and walleye, with average trophic position of 4.2, were classified as top predators (Fig. S4).

## Data analyses

We performed all data analyses using R v4.3.1 (R Core Team 2023) and the interface R Studio v2024.04.2 (R Studio Team 2024). We employed three linear models to assess the relationships between baseline $\delta^{13}$C values and DOC concentrations across the studied regions. The models included

**Fig. 2.** Lake mean benthic (orange) and pelagic (green) baseline $\delta^{13}C$ in relation to dissolved organic carbon concentrations (DOC, mg/L) across the studied regions. Significant linear relationships are indicated with shaded areas representing 95% confidence intervals.






the pelagic baseline, benthic baseline, and the difference between pelagic and benthic baseline $\delta^{13}C$ values as response variables. In each model, we treated DOC as a continuous predictor, while region (Sweden, Germany, Northern Ontario, Northwest Territories, Michigan, and Alaska) was included as a categorical predictor to account for potential geographic variability in $\delta^{13}C$ values. We used the lm function of the base R package. We evaluated the assumptions of linearity, homoscedasticity, and normality of residuals by visually inspecting the residual plots, and all models satisfied these assumptions.

We employed generalized additive models (GAMs) to investigate the effects of DOC concentrations on the pelagic diet proportion and trophic position across fish species. GAMs provide a flexible regression framework that effectively captures both linear and nonlinear relationships through smooth functions (Wood 2017). We constructed a GAM for each species. All models shared a consistent structure, incorporating smooth terms for continuous predictors—length and DOC—with a maximum of between 5 and 10 basis functions and the default thin spline. Length was included in the model to account for potential ontogenetic shifts in diet. Categorical variables, such as lake and region, were included as random intercepts to account for potential variability among locations. However, for certain species observed in only one region, we excluded the random effect for region. To assess the effect of tissue type (muscle vs. whole-body) in yellow perch trophic position, we conducted analyses both with and without the whole-body samples. Since the analysis without the whole-body samples yielded the same non-significant effect of DOC on yellow perch trophic position (Table S1), we present the results using the combined dataset for yellow perch. We used the mgcv R package v1.9-1 (Wood 2011). We managed model complexity by checking the maximum number of basis functions (k) using the gam.check function. We assumed a Gaussian error distribution for all models and evaluated the assumptions of GAMs by investigating diagnostics with the appraise function from the gratia R package v0.9.0 (Simpson 2024). To visualize the results, we employed the draw function from the same package.

# Results

## High DOC alters freshwater fish resource use in both pelagic and benthic pathways

Lake mean benthic baseline $\delta^{13}C$ values declined with increasing DOC concentrations across the studied regions (estimate = −0.28, SE = 0.08, $t = -3.34$, $p = 0.0014$; Fig. 2). A similar trend was observed for the lake mean pelagic baseline $\delta^{13}C$ values, which also decreased with higher DOC concentrations (estimate = −0.19, SE = 0.05, $t = -3.64$, $p = 0.0005$; Fig. 2). The models for both benthic and pelagic baseline $\delta^{13}C$ values demonstrated good fit, with $R^2_{adj}$ values of 0.40 and 0.35, respectively ($p < 0.0001$). At higher DOC concentrations (>30 mg/L), the lake mean baseline $\delta^{13}C$ values for benthic and pelagic sources began to overlap. In contrast,

**Table 2.** Effects of dissolved organic carbon concentrations (DOC, mg/L) on the pelagic diet proportion of grouped top predators and mid-trophic level fish and the six fish species, using generalized additive models, considering factors such as fish length, lake, and region.

| | Models | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Terms | Mid-trophic level fish | Top predators | European perch | Yellow perch | Black bass | Ninespine stickleback | Northern pike | Walleye |
| Intercept | 0.25(0.09) | 0.49 (0.09) | 0.31 (0.06) | 0.58 (0.08) | 0.23 (0.25) | 0.11 (0.05) | 0.39 (0.10) | 0.57 (0.04) |
| | $p = 0.0102$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p = 0.37$ | $p = 0.0308$ | $p < 0.0001$ | $p < 0.0001$ |
| s(DOC) | edf = 1.00 | edf = 3.47 | edf = 1.00 | edf = 2.63 | edf = 2.93 | edf = 1.85 | edf = 2.39 | edf = 3.57 |
| | $F = 4.45$ | $F = 4.88$ | $F = 12.76$ | $F = 0.46$ | $F = 1.38$ | $F = 2.93$ | $F = 3.83$ | $F = 4.08$ |
| | $p = 0.035$ | $p = 0.0012$ | $p = 0.0004$ | $p = 0.7300$ | $p = 0.4893$ | $p = 0.0665$ | $p = 0.0153$ | $p = 0.0034$ |
| s(length) | edf = 3.49 | edf = 6.82 | edf = 3.27 | edf = 2.83 | edf = 1.00 | edf = 1.03 | edf = 6.61 | edf = 2.01 |
| | $F = 9.89$ | $F = 11.87$ | $F = 9.76$ | $F = 3.87$ | $F = 3.09$ | $F = 1.60$ | $F = 9.61$ | $F = 1.63$ |
| | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p = 0.0045$ | $p = 0.0801$ | $p = 0.2136$ | $p < 0.0001$ | $p = 0.1746$ |
| s(lake) | edf = 49.34 | edf = 37.64 | edf = 16.04 | edf = 15.36 | edf = 12.57 | edf = 2.11 | edf = 35.82 | edf = 27.89 |
| | $F = 98.88$ | $F = 46.81$ | $F = 58.10$ | $F = 16.80$ | $F = 31.43$ | $F = 4.74$ | $F = 24.11$ | $F = 42.46$ |
| | $p = 0.048$ | $p < 0.0001$ | $p = 0.0201$ | $p < 0.0001$ | $p < 0.0001$ | $p = 0.0002$ | $p < 0.0001$ | $p < 0.0001$ |
| s(region) | edf = 3.21 | edf = 0.74 | edf = 0.55 | – | edf = 0.90 | – | edf = 0.76 | – |
| | $F = 36809$ | $F = 3211.79$ | $F = 909.10$ | | $F = 5710.75$ | | $F = 1142.66$ | |
| | $p < 0.0001$ | $p = 0.0472$ | $p = 0.1379$ | | $p = 0.0014$ | | $p = 0.0416$ | |
| Adjusted $R^2$ | 0.75 | 0.69 | 0.65 | 0.51 | 0.81 | 0.75 | 0.64 | 0.81 |
| Deviance explained | 75.5% | 70.1% | 66.2% | 53% | 81.8% | 75.8% | 65.8% | 81.7% |
| *n* | 1910 | 1518 | 925 | 519 | 271 | 195 | 832 | 686 |

**Note:** Statistical significance is indicated by $p < 0.05$. For the intercept, the estimate is reported with the standard error in parentheses. For smooth terms, the edf (effective degrees of freedom) and *F* value are reported.

we did not detect any relationship between the difference in pelagic and benthic baseline $\delta^{13}$C values and DOC concentrations (Estimate = −0.09, SE = 0.07, $t = 1.25$, $p = 0.2169$; Fig. S5). These results suggest that higher DOC concentrations, linked to increased terrestrial carbon input, likely enhance allochthonous carbon input into both benthic and pelagic pathways, as reflected in the more negative isotopic signatures of both baseline organisms.

## Species differences in pelagic diet proportion and trophic position of fish

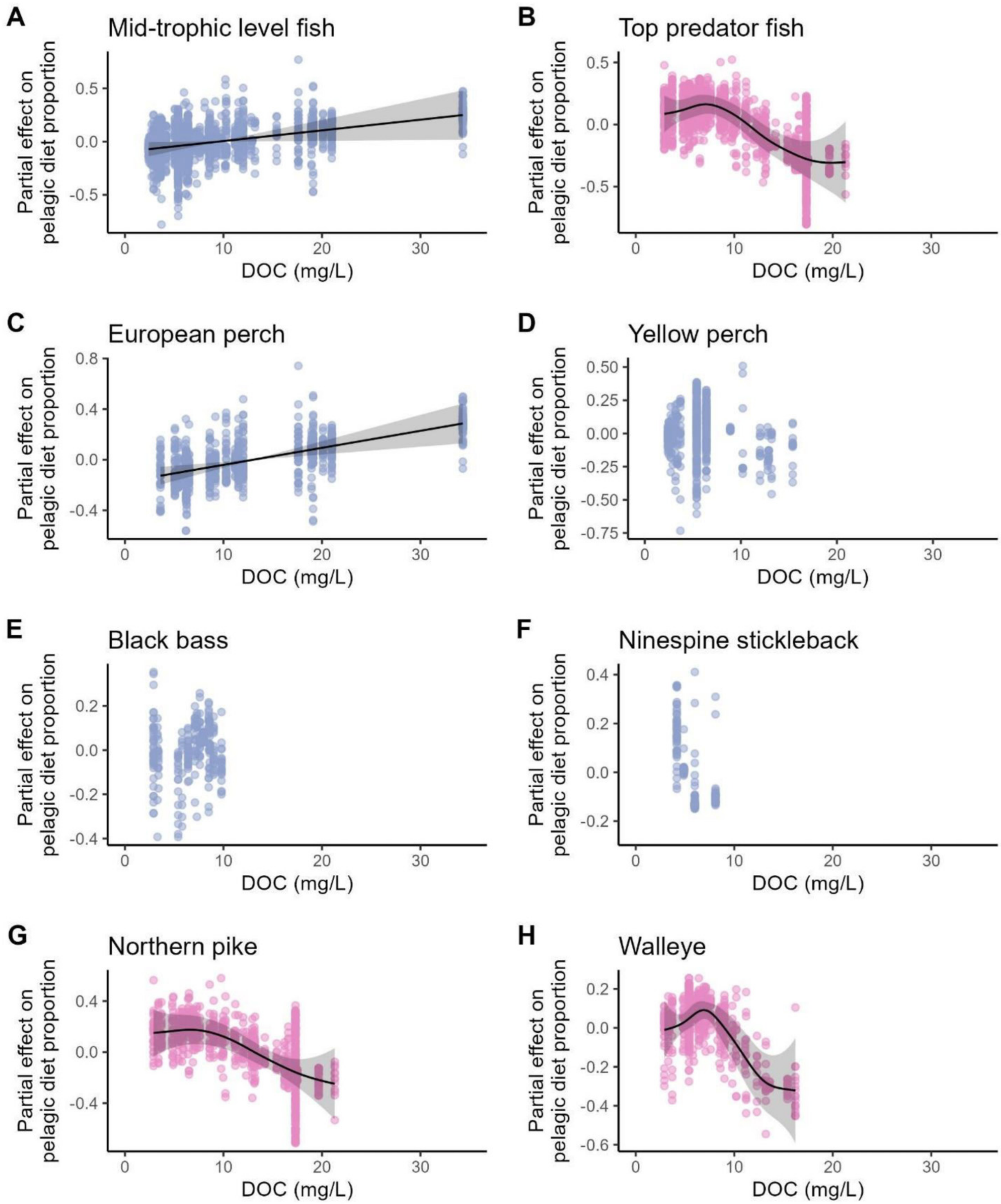
The proportion of pelagic diet varied with DOC for both mid-trophic level fish and top predators (Table 2; Fig. 3AB). The response of mid-trophic level fish to DOC was only observed for European perch, whose pelagic diet proportion increased linearly with more elevated DOC concentrations (Table 2; Fig. 3C). In contrast, no effects of DOC were detected for the other mid-trophic level fish species (Table 2), which include yellow perch (Fig. 3D), black bass (Fig. 3E), and ninespine stickleback (Fig. 3F). For top predator fish species northern pike and walleye, the pelagic diet proportions of northern pike and walleye exhibited nonlinear responses to DOC (Table 2; Fig. 3GH). For northern pike, the pelagic diet proportion remained unchanged at DOC levels up to 10 mg/L, after which it began to decline as DOC increased. Walleye had an initial increase in pelagic diet proportion at DOC levels up to approximately 8 mg/L, followed by a decrease as DOC concentrations increased.

No effects of DOC were observed on the trophic position of almost all fish species, regardless if they were a mid-trophic level fish or top predator fish (Table 3; Fig. 4AB). The exception was walleye for which trophic position increased linearly with rising DOC concentrations (Table 3; Fig. 4H). We detected a significant effect of total length on fish trophic position across all species (Table 3). Raw data for pelagic diet proportion and trophic position in relation to DOC concentrations are available in the supplementary material, shown in Figs. S6 and S7, respectively.

# Discussion

Our findings support the idea that rising DOC concentrations can enhance allochthonous inputs in food webs, fuelling both benthic and pelagic energy pathways. These changes can cascade through aquatic food webs, ultimately affecting top consumers, such as fish. Both benthic and pelagic baseline $\delta^{13}$C values decreased with increasing DOC concentrations; however, their difference remained relatively consistent across the gradient, with values beginning to overlap at higher DOC concentrations (∼30 mg/L). This likely indicates the growing importance of terrestrial-derived carbon in the food web across the studied regions. Fish responses to elevated DOC concentrations were species-specific, with distinct differences in pelagic diet proportions between European perch and top predator fish. European perch exhibited a linear increase in pelagic diet proportion with rising DOC, whereas northern pike and walleye

**Fig. 3.** Effects of dissolved organic carbon concentrations (DOC, mg/L) on the pelagic diet proportion of mid-trophic level fish (A), top predator fish (B), European perch (C), yellow perch (D), black bass (E), ninespine stickleback (F), northern pike (G), and walleye (H) using generalized additive models. All mid-trophic level fish are shown in blue, and all top predator fish in pink. Significant relationships are indicated with shaded areas representing 95% confidence intervals. Full models fit are provided in Table 2.






showed nonlinear patterns, with a decline in pelagic diet proportion at DOC concentrations above approximately 8–10 mg/L. Only walleye showed an increase in trophic position with increasing DOC concentrations. Our findings are important for predicting how environmental changes influence food web structure and impact top predators, particularly in ecosystems experiencing regional shifts in DOC concentrations.



Can. J. Fish. Aquat. Sci. **82:** 1–15 (2025) | dx.doi.org/10.1139/cjfas-2024-0343

**Table 3.** Effects of dissolved organic carbon concentrations (DOC, mg/L) on the trophic position of six fish species and grouped top predators and mid-trophic level fish, using generalized additive models, considering factors such as fish length, lake, and region.

| | Models | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Terms | Mid-trophic level fish | Top predators | European perch | Yellow perch | Black bass | Ninespine stickleback | Northern pike | Walleye |
| Intercept | 3.78 (0.03) | 4.18 (0.03) | 3.57 (0.09) | 3.51 (0.09) | 4.03 (0.11) | 4.02 (0.14) | 4.21 (0.04) | 4.14 (0.03) |
| | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ |
| s(DOC) | edf = 6.50 | edf = 1.00 | edf = 1.00 | Edf = 5.65 | edf = 1.00 | edf = 1.00 | edf = 1.36 | edf = 1.00 |
| | $F = 0.57$ | $F = 0.22$ | $F = 1.43$ | $F = 1.28$ | $F = 3.65$ | $F = 3.42$ | $F = 0.53$ | $F = 7.63$ |
| | $p = 0.7004$ | $p = 0.639$ | $p = 0.232$ | $p = 0.316$ | $p = 0.0571$ | $p = 0.0659$ | $p = 0.451$ | $p = 0.0059$ |
| s(length) | edf = 7.50 | edf = 3.80 | edf = 4.70 | edf = 1.38 | edf = 2.62 | edf = 2.38 | edf = 2.95 | edf = 2.51 |
| | $F = 153.34$ | $F = 227.89$ | $F = 110.19$ | $F = 277.29$ | $F = 18.45$ | $F = 3.67$ | $F = 196.95$ | $F = 280.64$ |
| | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p = 0.0136$ | $p < 0.0001$ | $p < 0.0001$ |
| s(lake) | edf = 44.65 | edf = 3.86 | edf = 1.69 | edf = 11.89 | edf = 13.34 | edf = 2.94 | edf = 3.71 | edf = 27.27 |
| | $F = 51.55$ | $F = 18.77$ | $F = 164.33$ | $F = 5.64$ | $F = 17.09$ | $F = 42.49$ | $F = 15.11$ | $F = -14.73$ |
| | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ | $p < 0.0001$ |
| s(region) | edf = 3.34 | edf = 0.00 | edf = 0.00 | – | edf = 0.61 | – | edf = 0.00 | – |
| | $F = 7092.34$ | $F = 0.00$ | $F = 0.00$ | | $F = 99.32$ | | $F = 0.00$ | |
| | $p = 0.0405$ | $p = 0.926$ | $p = 0.498$ | | $p = 0.1084$ | | $p = 0.378$ | |
| Adjusted $R^2$ | 0.79 | 0.56 | 0.80 | 0.70 | 0.61 | 0.62 | 0.64 | 0.66 |
| Deviance explained | 79.7% | 57.4 | 80.5% | 70.8% | 63.7% | 63.2% | 65.6% | 67.3% |
| $n$ | 1910 | 1518 | 925 | 519 | 271 | 195 | 832 | 686 |

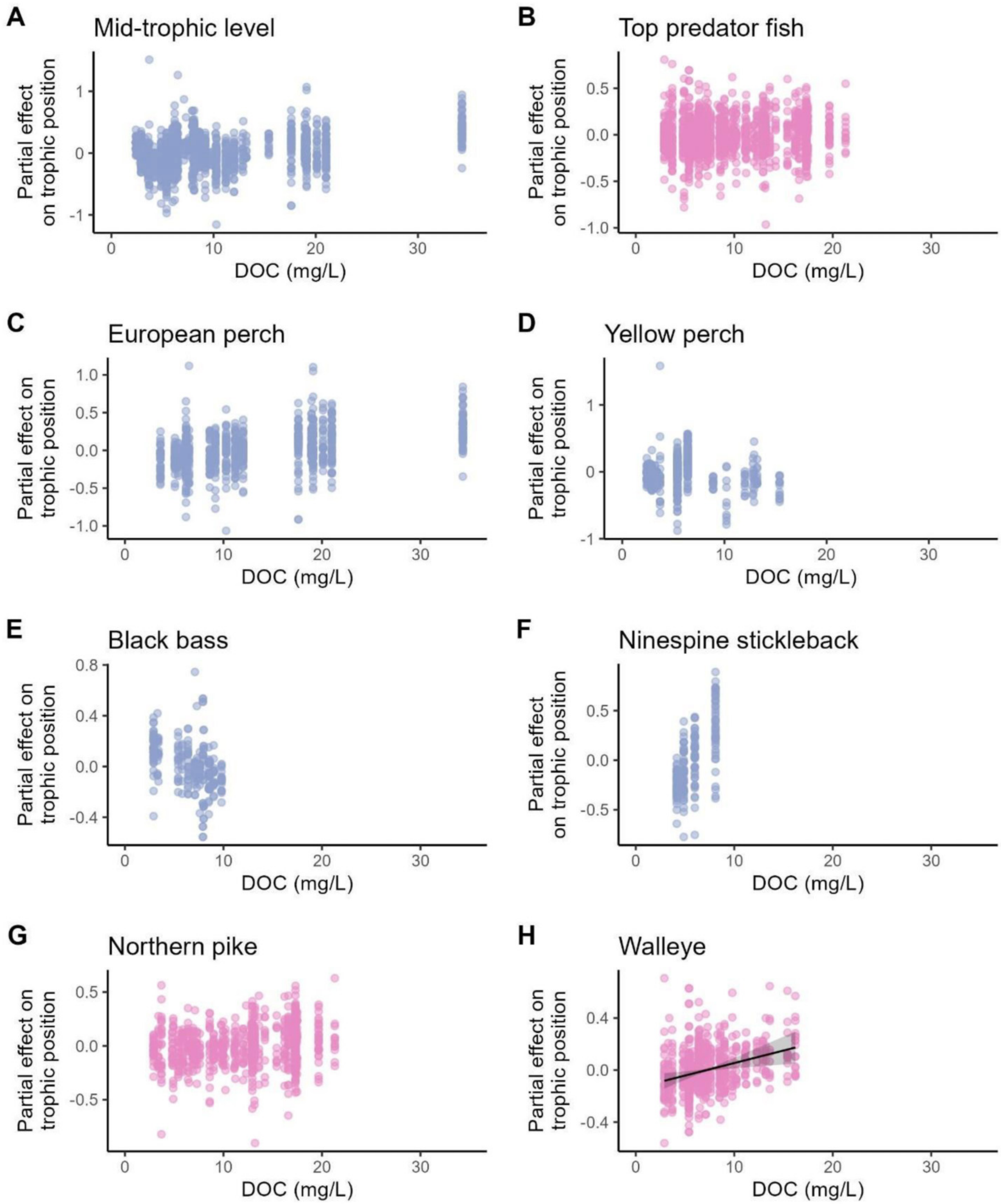
**Note:** Statistical significance is indicated by $p < 0.05$. For the intercept, the estimate is reported with the standard error in parentheses. For smooth terms, the edf (effective degrees of freedom) and $F$ value are reported.

## High DOC alters freshwater fish resource use in both pelagic and benthic pathways

The observed decrease in $\delta^{13}$C values in benthic baselines with increasing DOC concentrations may be attributed to a shift in the carbon sources supporting the benthic energy pathway. Typically, macroinvertebrates exhibit less negative $\delta^{13}$C values than pelagic organisms (Post 2002). A decline in $\delta^{13}$C could indicate a greater reliance on terrestrial or pelagic carbon by benthic macroinvertebrates. In moderate-DOC lakes, benthic primary production is often reduced, while pelagic primary production may increase (Karlsson et al. 2009; Seekell et al. 2015). Consequently, benthic invertebrates can rely heavily on a food web sustained by phytoplankton that settles to the lakebed (Goedkoop and Johnson 1996; Vadeboncoeur et al. 2003; Kathol et al. 2011). This aligns with the observed trend in our study, where $\delta^{13}$C values tend to range around −28 to −30‰ in high-DOC lakes, compared to −15 to −25‰ in low DOC lakes. Similar findings, where benthic organisms shift towards pelagic carbon sources, were also reported in Finnish freshwater lakes with increasing temperature and productivity levels (Hayden et al. 2019).

While this explanation is plausible for the benthic pathway, it does not fully account for the similar $\delta^{13}$C trends observed in the pelagic baselines with increased DOC concentrations. We propose that our findings are better explained by recent other research that supports that both benthic and pelagic food webs can use terrestrially derived carbon (Solomon et al. 2011; Brett et al. 2017; Tanentzap et al. 2017). High-DOC lakes typically receive substantial inputs of terrestrial organic matter, which both benthic and pelagic consumers increasingly rely on. This is consistent with our data, as terrestrial organic matter often has $\delta^{13}$C values around −28‰ in lakes (Wilkinson et al. 2013). Previous studies have demonstrated that both benthic and pelagic invertebrates can depend significantly on terrestrial-derived carbon, even in lakes with DOC concentrations as low as 3.7 mg/L (Solomon et al. 2008, 2011; Tanentzap et al. 2017). Primary consumers may use terrestrial carbon directly or indirectly through the consumption of heterotrophic bacteria (Brett et al. 2017). Because zooplankton can access terrestrial carbon sources via the microbial loop, increased DOC have been associated with lower $\delta^{13}$C values in zooplankton (Carpenter et al. 2005; Karlsson et al. 2009). Moreover, in the higher DOC lakes with $\delta^{13}$C values approaching −28 to −36 ‰, especially for pelagic baseline organisms, it is possible that there is an increasing reliance on methanotrophic food chains (Grey 2016), but this interpretation requires further validation. The lack of a significant relationship between the difference in pelagic and benthic baseline $\delta^{13}$C and DOC concentrations in our study suggests that both pathways are supported by allochthonous carbon. This further supports the hypothesis that terrestrial organic matter contributes to both food web pathways. At high DOC concentrations (∼30 mg/L), the emerging overlap between benthic and pelagic baseline $\delta^{13}$C values may suggest less distinct trophic pathways and a greater terrestrial-driven energy flow through-

**Fig. 4.** Effects of dissolved organic carbon concentrations (DOC, mg/L) on the trophic position of mid-trophic level fish (A), top predator fish (B), European perch (C), yellow perch (D), black bass (E), ninespine stickleback (F), northern pike (G), and walleye (H) using generalized additive models. All mid-trophic level fish are shown in blue, and all top predator fish in pink. Significant relationships are indicated with shaded areas representing 95% confidence intervals. Full models fit are provided in Table 3.




out the lake food web. Our findings add to the growing body of evidence emphasizing the critical role of terrestrial-derived carbon in freshwater food webs across broad spatial scales.

## Pelagic diet shifts with DOC in European perch, not other mid-trophic level fish

Mid-trophic level fish increased linearly in their pelagic diet proportion with rising DOC concentrations. The observed response was only explained by European perch in Swedish lakes, where the pelagic diet proportion increased linearly with DOC concentrations. This result aligns with previous studies in other Swedish lakes, which documented a decline in littoral contributions to fish diets as water transparency decreased (Bartels et al. 2012) and DOC levels increased (Bartels et al. 2016). The shift in European perch diet may reflect an increase in the proportion of pelagic resources, possibly driven by higher pelagic primary production relative to benthic production, as light attenuation intensifies in high-DOC lakes (Karlsson et al. 2009; Ask et al. 2009; Seekell et al. 2015).

In contrast, other mid-trophic level fish such as yellow perch, black bass (smallmouth and largemouth), and ninespine stickleback did not exhibit significant shifts in pelagic diet proportions with increasing DOC. Given the ecological similarities between yellow perch and European perch, we expected a comparable response to DOC, but this was not detected. Previous research on yellow perch has primarily focused on the effects of DOC on growth rates (e.g., Benoît et al. 2016) and the effect of water clarity on diet and foraging behaviour, with mixed results (Hansen et al. 2022). Our findings differ from Stasko et al. (2015), who reported an expansion of the isotopic niche of smallmouth bass and increased pelagic feeding in response to decreasing water clarity. However, caution is necessary when comparing results, as Stasko et al. (2015) did not account for variations in stable isotope baselines and focused on water clarity rather than DOC specifically. Past studies have shown that largemouth bass can adjust to increased DOC concentrations by altering their habitat use and prey selectivity (Schindler and Gunn 2003; Koizumi et al. 2018). The lack of consistent DOC effect across mid-trophic level fish species highlights the complexity of DOC-driven food web changes. Shifts in resource use are not uniform across taxa and may depend on species-specific ecological traits, prey availability, and local habitat conditions. Furthermore, compared to the Swedish lakes in which European perch showed a positive, linear response, the narrower range of DOC concentrations in lakes with yellow perch, black bass, and ninespine stickleback may have limited our ability to detect DOC-related effects on pelagic diet proportion in these mid-trophic level fish species. These results emphasize the need to consider interactions across multiple trophic levels and energy pathways when assessing the ecological consequences of changing DOC levels.

## Non-linear pelagic diet shifts in top fish predators with increasing DOC

The two top predator species, walleye and northern pike, exhibited similar nonlinear shifts in pelagic diet proportions in response to DOC. Walleye showed an increased in pelagic diet proportion up to a DOC concentration of 8 mg/L, after which it declined. Northern pike maintained a stable pelagic diet proportion until 10 mg/L, followed by a decrease at higher DOC concentrations. These parallel trends suggest a general reduction in pelagic diet proportion for both top predator fish species at high DOC concentrations. One possible explanation is that moderate DOC concentrations may initially enhance pelagic productivity and associated prey, particularly benefiting walleye, which are known to forage more effectively in low-light environments (Hartman 2009; Harvey 2009). Northern pike, as ambush predators, depend on clear visual conditions and may be unable to exploit pelagic prey under higher DOC concentrations (Harvey 2009). It has been suggested that predators targeting larger prey are more vulnerable to reduced visibility compared to those that feed on smaller prey (Utne-Palm 2002; De Robertis et al. 2003). This is because encounter rates of larger prey depend on longer detection distances, while the negative effect of browning (or turbidity) on detection distance is exponential (Jönsson et al. 2013). At higher DOC concentrations, it is possible that both top predator fish species shifted towards benthic prey, possibly relying on more tactile or vibration cues. While these mechanisms were not directly tested in our study, they provide plausible ecological explanations for the observed patterns.

Our findings differ from previous research that found no significant changes in walleye resource use with respect to water clarity (Edmunds et al. 2019; Nanayakkara et al. 2021) or DOC concentrations (Stasko et al. 2015). However, Jarvis et al. (2020) reported a nonlinear relationship between DOC and walleye production in Ontario lakes, highlighting the complexity of these interactions. Similarly, Moslemi-Aqdam et al. (2021) observed slower growth rates and more negative $\delta^{13}C$ ratios in northern pike from lakes with high DOC concentrations in the Northwest Territories. These findings suggest that dietary shifts in response to DOC, as observed in our study, may have important implications for fish production in freshwater ecosystems, particularly for top predator species such as walleye and northern pike.

## No effect of DOC on trophic position in most freshwater fish species

While DOC influenced pelagic diet proportions in some fish species, it showed no clear relationship with trophic position for most species. The exception was walleye for which we observed an increase in trophic position at more elevated DOC concentrations. Although pelagic energy sources can support smaller organisms than nearshore benthic energy (Maitland et al. 2024), trophic position is often more strongly correlated with body size, a key factor in food web structure (Petchey et al. 2008; Romanuk et al. 2011). Our results support this, as total length was a significant predictor of trophic position across all species. Dietary shifts do not always translate to changes in trophic position, especially if fish continue feeding within the same trophic level (e.g., Charette and Derry 2024). Additionally, trophic shifts have been associated with prey availability and ecosystem size (Eloranta et al. 2015), further complicating the relationship between DOC and trophic position. Overall, our findings suggest that while DOC can alter energy pathways in food webs and dietary composition in some fish species, its influence on fish trophic position is relatively limited.

## Limitations

Differences in lake characteristics, such as lake size, stratification, and water residence time, can all mediate fish responses to DOC changes (Kelly et al. 2018). Although our study focused on DOC as a key driver of basal resources and freshwater fish dietary niches, other correlated environmental factors, such as temperature, light absorption, and nutrient availability, may play a role (Seekell et al. 2015; Solomon et al. 2015; Creed et al. 2018). Inconsistencies in our results may reflect different aspects of browning, such as its effects on fish visual environment or basal resources. Moreover, DOC and water colour are not always tightly correlated (Rodríguez-Cardona et al. 2023), adding further complexity. Future research could disentangle the effects of different abiotic and biotic factors for gaining a more comprehensive understanding of how DOC and related environmental variables shape fish trophic ecology. Stable isotope analysis is a powerful tool for studying trophic pathways, but its accuracy depends on well-defined isotopic baseline signatures (Post 2002). Overlap in baseline $\delta^{13}$C and $\delta^{15}$N values, along with variability in trophic discrimination factors, can reduce the precision of dietary or trophic position inferences (Kjeldgaard et al. 2021). Future studies should ideally include complementary methods such as gut content or fatty acid analyses to strengthen conclusions.

# Conclusion

Our study highlights the important role of terrestrial-derived carbon in supporting both pelagic and benthic food web pathways and affecting food web structures in freshwater ecosystems. While DOC influenced dietary shift in some freshwater fish species, most showed no clear relationship between DOC and trophic position, reflecting the complexity of these interactions. Our findings suggest that DOC-driven changes in food web structure are not uniform but may be species-specific depending on ecological traits. These findings enhance our understanding of how regional shifts in DOC concentrations impact food web structure and energy flow, with potential consequences for freshwater fish species of recreational and commercial importance. Understanding these interactions is key to managing freshwater ecosystems in the face of ongoing environmental changes.

# Acknowledgements

This contribution was made possible by a working group grant, led by AMR, from the Groupe de Recherche Interuniversitaire en Limnologie/the Interuniversity Research Group in Limnology (GRIL), which is funded in turn by the Fonds de recherche du Québec—Nature et Technologie (FRQNT). Data from Michigan, and work by CTS on this project, were supported by the U.S. National Science Foundation under awards 1754363 and 1754561. Data from the Northwest Territories were generated with support from the Canada First Excellence Research Fund (Global Water Futures; Northern Water Futures project), the Cumulative Impacts Monitoring Program (Government of the Northwest Territories), the Northern Contaminants Program, Dehcho First Nations, and Dehcho Aboriginal Aquatic Resources and Oceans Management program. Data from Alaska were made possible by funding from the Arctic Landscape Conservation Cooperative (PI: Christian Zimmerman). Data from Sweden and Germany and work by MA and PE on this project were supported by the Swedish Research Council Formas, grant number: Dnr. 942-2045-365. Data from northern Ontario waters was generated through research supported by the Ontario Ministry of Natural Resources, the Ontario Ministry of Environment, Conservation and Parks, and the Natural Sciences and Engineering Research Council of Canada.

# Article information

## History dates

Received: 8 November 2024
Accepted: 21 July 2025
Accepted manuscript online: 20 August 2025
Version of record online: 2 October 2025

## Notes

This paper is part of a Collection entitled Freshwater browning and coastal darkening: effects and processes.

## Copyright



## Data availability

The data underlying this article are available in the Dryad Digital Repository at doi:https://doi.org/10.5061/dryad.6djh9w1bt.

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## Competing interests

The authors declare there are no competing interests.

## Supplementary material

Supplementary data are available with the article at https://doi.org/10.1139/cjfas-2024-0343.

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